Review: Probability and Statistics

CS3263

Foundations of Artificial Intelligence

Topics

- Probability Basics
- Conditional Probability and Independence
- Random Variables
- Probabilistic Models
- Statistical Concepts

A Little Background …

- Probability theory was first developed more than 300 years ago, motivated by games of chance
  - Cardano, Fermat, Pascal, Bernoulli, Leibniz, Bayes, De Moivre,
  - Laplace, Gauss, Poisson, Chebyshev, Markov, …
- Wide applications in modern days
  - Games
  - Biology and medicine
  - Statistical Analysis
  - Computer Science
    - Complexity analysis of algorithms
    - Uncertainty management
    - Machine learning
    - Robotics
    - Adaptive systems

Interpretations of Probability

- Frequency Interpretation
  - Applied to statistical theory for analysis of historical data
  - Fraction or proportion of an event
    - Roll a die many times and count the no. of times that “Head” appears
  - Odds of an event
    - The odds of getting a 3 are one to five in rolling a die
- Measure Approach
  - Measures the event of interest
  - Indicating the “size” of an event on a scale where an event that is certain to occur measures

Interpretations of Probability

- Subjective View
  - Subjective belief on the chance that a future uncertain event would happen
  - Captures judgment of decision maker
  - Used when human behavior and/or competitive forces are involved that could distort historic averages

Quantifying Uncertainty with Probability

- It is likely to rain today
  - “Likely” means:
  - 50 % 60% 80% chance?
- There is a non-negligible chance that you will fail CS4246
  - “Non-negligible” means:
  - less than 5 % 10% 5-10% chance?
- Differentiate among “probable”, “remote”, “reasonably possible”
  - Probable: likely to occur
  - Remote: slight chance
  - Reasonably possible:
    chance of occurring is somewhere between slight and likely

Basic Definitions

- Experiment
  - A procedure or an operation that yields one of a given set of possible outcomes
- Outcome
  - A specific result of an uncertain experiment
- Sample space
  - The set of all possible outcomes of an experiment
- Event
  - A subset of the sample space
  - A single outcome is also an event

Experiments, Outcomes, and Events

Experiment	Outcome	Sample Space
1. Toss a coin	Obtain a head	S = { H, T}
2. Roll a die twice	Obtain 3 then 5	S = { (i, j): i =1, ... 6, =1,...,6 }
3. Take CS5340	Obtain a B grade	S = { A+, A, ..., D+, D, F }

Experiment	Event A	Relation to S
1. Toss a coin	Obtain a head or tail	A = S
2. Roll a die twice	Obtain total of 8	A = { (6,2),(5,3),(4,4),(3,5),(2,6) }
3. Take CS5340	Obtain a passing grade	A = { A+, A, ..., D+, D }

Defining Probabilities

- Let A and B be any two events in a finite sample space S of equally likely outcomes
- The probability that event A (or B) will occur is

$$P(A) = |A| / |S|; \quad 0 \le P(A) \le 1$$

- The probability that event A (or B) does not occur is

$$P(A^c) = 1 - P(A)$$

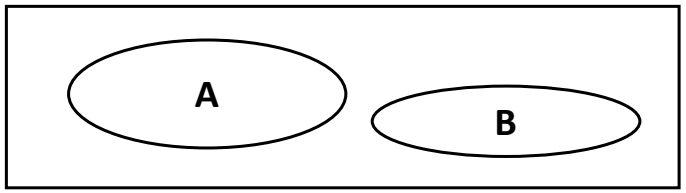
- The probability that either A or B occurs is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

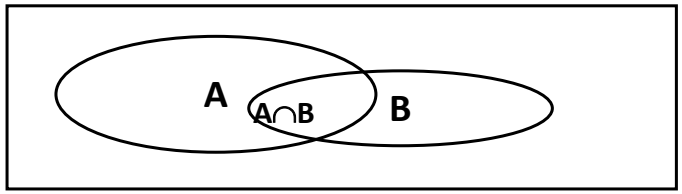
Defining Probabilities

- Events A and B are mutually exclusive
  - if they cannot occur simultaneously in the same performance of an experiment or operation; and
  - the additive law of probability states that the probability that either A or B occur is

$$P(A \cup B) = P(A) + P(B)$$



A and B mutually exclusive



A and B not mutually exclusive

Probability Distributions

- Let S be the sample space of an experiment with a finite or countable number of outcomes

- The function P from the set of all events of the sample space S (or $\mathcal{P}(S)$) to [0, 1] that conforms to the axioms of probability is called a probability distribution

Axioms of Probability

- Assumptions:
  - An event A is a subset of the sample space S
  - $P(A)$ represents the probability of event A

- Axioms of probability
  1. (Nonnegativity) $P(A) \geq 0$ for any event A
  2. (Normalization) $P(S) = 1$ and $P(\emptyset) = 0$
  3. (Additivity) If events A_i, $i = 1,2,\ldots$ are mutually exclusive, then $P(\cup_i A_i) = \Sigma_i P(A_i)$

Conditional Probability and Independence

Conditional Probability

- Let A and B be any two events
  - Assuming that $P(A) > 0$, the conditional probability that B occurs, given that A has occurred is:

$$P(B|A) = P(A \cap B) / P(A)$$

  - This is the probability that B occurs, given that A is observed
  - Out of the total probability of the elements of A, $P(B|A)$ is the fraction that is assigned to possible outcomes that also belong to B

Example: Family Planning

- Given that a family with two children has at least one girl:
  - What is the probability that both children are girls?
  - Assume that any child born is a boy or a girl
- Answer: 1/2? 1/3? 1/4? Others?

Example: Family Planning

- Let A be the event "at least one child is a girl"
- Let B be the event "both children are girls"
- $P(A) = \mathbf{3/4}$
- $P(B) = \mathbf{1/4}$
- $P(A \cap B) = \mathbf{1/4}$
- $P(B|A) = P(A \cap B) / P(A) = \mathbf{1/4 * 4/3 = 1/3}$

First child: Girl

Second child: Girl

(G, G)	**(G, B)**
(B, G)	**(B, B)**

Second child: Boy

First child: Boy

Alternate Form

- Rewriting the definition for conditional probability:
  - The probability that both A and B occur is:

$$P(A \cap B) = P(B|A)P(A) = P(A|B)P(B)$$

  - The probability that A and B both occur is the probability that B (A) occurs times the probability that A (B) occurs, given that B (A) has occurred.

The Chain Rule

- Extending to multiple events:

$$P(A_1 \cap A_2 \cap \ldots \cap A_k)$$

$$= P(A_k \mid A_1 \cap A_2 \cap \ldots \cap A_{k-1}) \ldots P(A_3 \mid A_1 \cap A_2)\, P(A_2 \mid A_1)\, P(A_1)$$

$$= P(A_1)\, P(A_2 \mid A_1)\, P(A_3 \mid A_1 \cap A_2) \ldots P(A_k \mid A_1 \cap A_2 \cap \ldots \cap A_{k-1})$$

Independence

- If A and B are independent, then the probability that both A and B occurs is:

$$P(A \cap B) = P(A)P(B)$$

- If two events that can occur with positive probability are mutually exclusive, then they must be dependent:

  If A occurs, then B cannot occur, and

$$P(A \cap B) = 0$$

- Events $A_1, A_2, \ldots, A_n$ in a sample space S are mutually independent iff the probability of the intersection of any subset of the events is the product of the probabilities of the events in the subset.

Conditional Independence

- Let A, B, and C be any events.
  - Given C, A and B are called conditionally independent if:

$$P(A \cap B \mid C) = P(A \mid C)P(B \mid C)$$

  - Or, equivalently:

$$P(A \mid B \cap C) = P(A \mid C)$$
$$P(B \mid A \cap C) = P(B \mid C)$$

- Note:
  - Independence does not imply conditional independence, and vice versa!

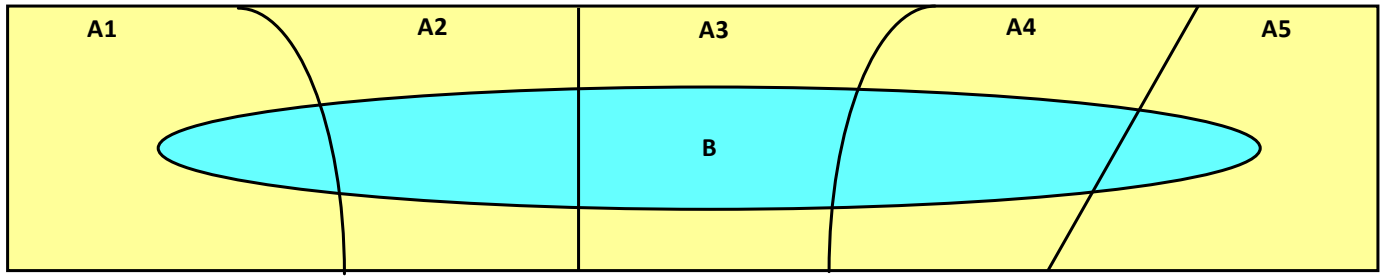
The Law of Total Probability

- A partition of the sample space S is any collection of mutually exclusive events in S whose union is S
- Let $\{A_i : i = 1, 2, \ldots, n\}$ be n subsets of S. This collection forms a partition of S if and only if

$$A_i \cap A_j = \varnothing \qquad (\text{when } i \neq j) \qquad \text{and} \qquad \cup_i A_i = S$$

- The law of total probability:

$$P(B) = \Sigma_i \; P(A_i \cap B) \quad \text{or} \quad P(B) = \Sigma_i \; P(B \mid A_i)\, P(A_i)$$



Mutually Disjoint Events and Independence

- Let A and B be events in a sample space S
- Suppose $A \cap B = \varnothing$, $P(A) \neq 0$, and $P(B) \neq 0$
- Show that $P(A \cap B) \neq P(A) \cdot P(B)$

- Solution:
  - Because $A \cap B = \varnothing$, $P(A \cap B) = 0$ by Axiom 2
  - But $P(A) \cdot P(B) \neq 0$ because neither $P(A)$ nor $P(B)$ equals zero
  - Thus $P(A \cap B) \neq P(A) \cdot P(B)$

Baye's Theorem

- Let A and B be any two events
  - Recall:

$$P(A \cap B) = P(A|B)P(B)$$

  - Hence:

$$P(A|B) = P(A \cap B) / P(B)$$

- Baye's Rule:

$$P(A|B) = P(B|A)P(A) / P(B)$$

  - The marginal pmf P(A) is called the prior of A
  - The conditional pmf P(A|B) is called the posterior of A in the light of the new information or evidence provided by knowing B
  - The conditional pmf P(B|A) is called the likelihood of B given A
  - The marginal pmf P(B) is a normalizing factor

- In general

$$P(B_i|A) = P(A|B_i)P(B_i) / \Sigma_i P(A|B_i) P(B_i)$$

Bayes' Rule in Action

- In a diagnosis problem, Baye's Rule often written as:

$$P(H|e)= P(e|H)P(H)/ P(e)$$

  - People often find $P(H|e)$ hard to assess
    E.g., What is the probability of pneumonia given fever?
- On the other hand:
  - $P(e|H)$, the probability of seeing the outcome given that the hypothesis is true, can usually be confidently assessed
    E.g., There is a 80% that a patient with pneumonia will have fever

Coherence

- Coherence means consistency
- The laws of probability ensure consistency in the mathematical analysis
  - Probabilities must be between 0 and 1
  - Probabilities must add up to 1 over all possible outcomes
  - Joint distributions, conditional distributions, and marginal distributions must be consistent

Example

- There is
  - a 40% of rain today,
  - a 30% chance of rain tomorrow, and
  - a 20% chance that it will not rain on either day
- Let A = “it rains today”, B = “it rains tomorrow”

  $P(A) = 0.4$ $P(B) = 0.3$ $1 - P(A \cup B) = 0.2$

- What is $P(A \cap B)$?
- How to fix the inconsistency?

Random Variables

Random Variables

- A random number X is an uncertain quantity that may take on several values x derived from the set of all possible outcomes (an abstraction)
- X is a function from the sample space of an experiment to
  - A binary set – Boolean Random Variables
  - A finite, countable set – Discrete Random Variables
  - A set of real numbers – Continuous Random Variables
- Example: Number of chocolate chips in a cookie
  - $X = 1, 2, \ldots, n, n \in \mathbf{Z+}$
- Example: Tomorrow's maximum temperature
  - $T = [22, 38]$

Probability Distributions

- Set of probabilities associated with all possible outcomes of a random variable or uncertain quantity is called its probability distribution
- Discrete random variable
  - Probability mass function (pmf)

    $p(x) = P(X=x) = P(X \text{ takes on the value } x)$

    $\sum_{x \in X} p(x) = 1 \quad \text{and} \quad p(x) \geq 0$
  - Cumulative distribution function (cdf)

    $P(x) = P(X \leq x) = \sum_{x_i \leq x} p(x_i), \quad x \geq 0$
- Continuous random variable
  - Probability density function (pdf)

    $\int_{x \in X} f(x)\, dx = 1 \quad \text{and} \quad f(x) \geq 0$
  - Cumulative distribution function (cdf)

    $P(x) = Pr(X \leq x\} = \int^{x} f(u)\, du, \quad u \geq 0$

Marginal and Conditional Probabilities

- Let X and Y be two random variables
- Marginal pmf

  $p_X(x) = P(X = x), \qquad x \in X$

  $p_Y(y) = P(Y = y), \qquad y \in Y$
- Conditional probability of Y given X

  $p_{Y|X}(y \mid x) = P(Y = y \mid X = x), \quad x \in X, \; y \in Y$
- Conditional probability of X given Y

  $p_{X|Y}(x \mid y) = P(X = x \mid Y = y), \quad x \in X, \; y \in Y$
- The laws of probability apply to each of the above:
  - Each distribution is nonnegative
  - Conditional probabilities over the outcomes of the random variable of interest (not the conditioning event) sum to 1

Joint Probabilities

- Joint probability of two random variables X and Y

  $p_{X,Y}(x, y) = P(X = x, Y = y), \quad x \in X \text{ and } y \in Y$

  $p_{X,Y}(x, y) = p_{Y|X}(y \mid x)\, p_X(x) = p_{X|Y}(x \mid y)\, p_Y(y)$

- Marginalization
  - Process of summing over the unwanted variables
  - Given the joint probabilities or density $p_{X,Y}(x, y)$, compute the marginal probabilities or densities of X and Y by:

    $p_{X,}(x) = \sum_y p_{x,y}(x, y)$ and $p_{y,}(y) = \sum_x p_{x,y}(x, y)$

    $p_{X,}(x) = \int_y p_{x,y}(x, y)\, dy$ and $p_{y,}(y) = \int_x p_{x,y}(x, y)\, dx$

Bayes' Theorem

- Given two random variables X and Y

$$p_{Y|X}(y \mid x) = p_{X|Y}(x \mid y)\, p_Y(y) \,/\, p_X(x)$$

  - The marginal pmf $p_Y(y)$ is called the prior of Y
  - The conditional pmf $p_{Y|X}(y \mid x)$ is called the posterior of Y in the light of the new information or evidence provided by knowing $X = x$
  - The conditional pmf $p_{X|Y}(x \mid y)$ is called the likelihood of X given Y

Probabilistic Models

Types of Probabilistic Models

- Subjective probabilities
  - Subjective assessment of uncertainty in terms of probability
  - How to make such judgments?
  - What do such judgments imply?
- Theoretical probability models
  - Modeling uncertainty with theoretical probability distributions
  - Theoretical probability distributions have well-defined characteristics and useful statistics
- Deriving probabilities from data
  - Construct probability distributions from data
  - Fit theoretical probability models with data
  - Model relationships with data

Subjective Probabilities

- Necessary for rigorous approach to modeling uncertainty
  - e.g. Earthquake predictions; medical diagnosis, etc
- Subjective interpretation of probability
  - Probability represents an individual's degree of belief that a particular outcome will occur
  - Long-run frequency interpretation not always applicable
  - An event has occurred, but its outcome may still be unclear; uncertainty is in the mind, not the outcome

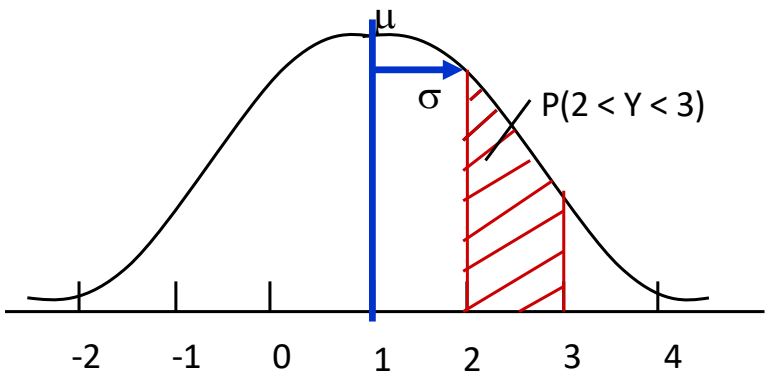
Theoretical Probabilistic Models

- Standard distributions for random variables
  - Discrete distributions
    - Uniform, binomial, Poisson
  - Continuous distributions
    - Normal (Gaussian), exponential
- Examples with appropriate standard distributions:
  - Education testing
  - Market research
  - Quality control
  - Election outcome prediction
  - Capacity planning
  - Environmental risk analysis

Normal (Gaussian) Distribution

- Distribution for real valued outcomes with density function:

$$f(y) = \frac{e^{-(y-\mu)^2/(2\sigma^2)}}{\sigma\sqrt{2\pi}}$$



- Where
  - E(Y) = μ is the mean and
  - Var(Y) = σ^2 is the variance
- Multivariate Gaussian Distribution
  - Generalization to Gaussian distribution for n real random variables
  - Represented by an n x 1 mean vector μ and a symmetric n x n covariance matrix Σ.

Using Theoretical Probability Models

- Approach 1:
  - Consider relevant characteristics of system
  - Determine if characteristics match assumptions in a standard distribution
  - Use standard distribution to generate probabilities
  - Assumption: physical system can be represented adequately in model
- Approach 2:
  - Assess a subjective probability distribution from data
  - Find a standard distribution that “fits” the subjective distribution to facilitate probability and expected-value calculations
  - Assumption: theoretical model adequately fits the assessed judgments

Using Data to Model Uncertainty

- Using empirical data to:
  - Construct probability distributions
  - Fit theoretical probability models
  - Model relationships among variables
- Assumptions
  - Data is sufficient to make reliable estimations
  - Data requirements depend on decision contexts
  - Data adequately reflects the uncertainty faced
  - Past data indicates what the future holds

Caveats

- Need sufficient data for reliable estimates
  - At least 5 in least likely category
- Do the probability estimates based on data truly reflect uncertainty in question?
  - Are the data appropriately collected?
- Does the past indicate what the future may hold?
  - Are the data relevant?
- Can subjective assessment do better?
  - Are the data comprehensive?

Statistical Concepts

Expectation or Expected Value

- Definition:
  - Probability-weighted average of possible values of a random variable
- Expectation for discrete random variable

$$E[X] = \sum x_i \, p(x_i)$$

- Expectation for continuous random variable

$$E[X] = \int x \, f(x) \, dx$$

- In general:

If $Y = g(X)$, then $E[Y] = E[g(X)]$

If $Y = a+bX$, then $E[Y] = a+bE[X]$

$$E[X+Y] = E[X] + E[Y]$$

Variance and Standard Deviation

- Measures of variability

- Variance

$$V(X) = E(X^2) - E(X)^2$$
$$V(X + Y) = V(X) + V(Y)$$

- Standard Deviation

$$\sigma(X) = V^{1/2}(X)$$

Joint Expectation

- Joint expected value
  - For discrete random variable

$$E[XY] = \sum_{x \in X} \sum_{y \in Y} xy\, p_{X,Y}(x, y)$$

  - For continuous random variable

$$E[XY] = \int_{x \in X} \int_{y \in Y} xy\, f_{X,Y}(x, y)$$

Covariance and Correlation

- Measures of dependence
- Covariance

$$Cov\,[XY] = E\,[XY] - E\,[X]\,E\,[Y]$$

- Correlation

$$\rho_{XY} = Cov(X,Y)/\,\sigma_X\,\sigma_Y$$

  - If $\rho_{XY} > 0$, the variables are positively correlated
    - As one increases the other increases
  - If $\rho_{XY} < 0$, the variables are negatively correlated
    - As one increases the other decreases

Linear Regression

- Given an independent variable X and a dependent variable Y such that:

$$y = \beta_0 + \beta_1 x + \varepsilon_x$$

- Where ε_x is a random variable dependent on value x of X.
- To find linear relationship from a random sample of n values of X and Y
  - Multiple linear regression – more independent variables $X_1, \ldots, X_n$
- To estimate values of β_0 and β_1, find values of parameters (coefficients) b_0 and b_1 that minimize:

$$\sum_{i=1}^{n} [y_i - (b_0 + b_1 x_i)]^2$$

Noise and Expectation

- ε_x is a random variable dependent on value x of X, with the following properties:
  - For every value x of X, ε_x is normally distributed with 0 mean
  - For every value x of X, ε_x has the same standard deviation σ
  - The random variables ε_x for all x are mutually independent
- The expected value of Y given a value x of X is given by:

$$E[Y \mid X = x] = \beta_0 + \beta_1 x + \varepsilon_x$$

- Note:
  - Expected value of Y is a deterministic linear function of x
  - Actual value y of Y is not uniquely determined by value of X because of random error ε_x

Standard Error of Coefficient

- Computes confidence in how close the approximations are to the true values of β_0 and β_1
  - Assumption: linear relationship exists
- Recall:
  - For normal distribution, 95% of mass falls in interval with endpoints being the mean ± 1.96 times standard deviation
- If σ_0 is the SE coefficient for b_0, can be 95% confident that:

$$\beta_0 \in (b_0 - 1.96\sigma_0, b_0 + 1.96\sigma_0)$$

Other Confidence Measures

- T-value:
  - T = coefficient / SE coefficient
  - Large value of T - more confident that estimate is close to the true value
- P-value:
  - If true value of parameter is 0, the probability of obtaining data more unlikely than the result

Coefficient of Determination

- R^2 = square of the sample correlation coefficient, with value between:
  - 0% - there is no linear dependence between sample values of X and Y
  - 100% there is perfect linear dependence between sample values of X and Y
- Large value of R^2 - more confident that there is really a linear relationship between X and Y

Hypothesis Testing

- Establish a null hypothesis H_0 that the experimental result is purely due to chance.
- The alternative hypothesis H_A or H_1 indicates that there are some positive or negative effects
- Take a random sample from the population
- If sample data are consistent with H_0:
  - Do not reject the null hypothesis
- If sample data are inconsistent with H_0 (in direction of H_A):
  - Reject the null hypothesis; and
  - Conclude that the alternative hypothesis is true

Hypothesis Testing and Confidence Level

- If hypothesis is that observations are from a normal distribution with mean μ and variance σ^2, reject hypothesis if sample mean m is outside the range $\mu \pm z\sigma/\sqrt{n}$ for the value z, which depends on the significance level
- For example, at significance level $\alpha=0.05$, reject hypothesis if sample mean m is not in the range $\mu \pm 1.96\sigma/\sqrt{n}$
- In determining whether the actual mean is larger (or smaller) than μ, reject the hypothesis at a significance level of $\alpha=0.025$ if $m>\mu+1.96\sigma/\sqrt{n}$
  (or $m < \mu - 1.96\sigma/\sqrt{n}$)
- $(1-\alpha)$% confidence interval for μ

  With probability $(1-\alpha)$,
  - $\mu - z\sigma/\sqrt{n} < m < \mu + z\sigma/\sqrt{n}$; OR
  - $m - z\sigma/\sqrt{n} < \mu < m + z\sigma/\sqrt{n}$

Approximation

- To do hypothesis testing or obtain confidence intervals using the Normal Distribution, need to know μ and σ.
- For large sample sizes (>30), usually quite accurate to replace mean and variance with sample mean and variance

$$\mu \approx m = \frac{1}{n}\sum_{i=1}^{n} y_i \qquad \sigma^2 \approx s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - m)^2$$

- According to the central limit theorem, for large enough sample, distribution of sample mean is approximately normal even if the original random variable is not normally distributed.

Example

- Shopping times of $n=64$ randomly selected customers at a local supermarket were recorded.
- Average and variance were 33 minutes and 256 respectively.
- Estimate μ the true average shopping time per customer, with a confidence coefficient of $(1-\alpha)=0.9$
- Use sample variance $s^2=256$ to approximate $\sigma^{2.}$
- Given $\sigma/\sqrt{n} \approx s/8=16/8=2$
- From the table (what table?), need $z = 1.64$ so
  - $m-1.64\sigma/\sqrt{n} < \mu < m+1.64\sigma/\sqrt{n}$
  - $33-3.28 < \mu < 33+3.28$
  - $29.72 < \mu < 36.28$

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